W1

Mathematical Modelling

Relation between parameters that are of influence in this real world problem

FUNCTIONS

→ For values in set A → there is only <u>ONE</u> value in set B
↓
1 output per input

$$\text{Domain } (x) \longrightarrow \underset{\text{Rule}}{f(x)} \longrightarrow \text{Codomain } (y)$$

describe functions → formula

POLINOMIALS

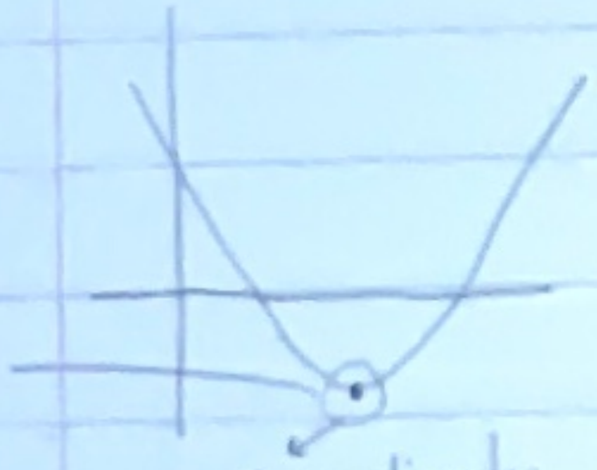
$f(x) = ax + b$ — intersection to y → y-intersect

a → pendiente (SLOPE)

Degree 2 → Quadratic

y-intersect

$y = ax^2 + bx + c$



VERTEX

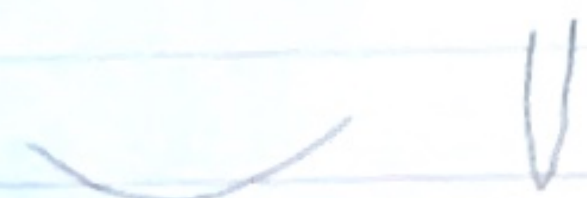
x coordinate: $\frac{-b}{2a}$

→ graph – parabola

→ a → orientación y axis

∪ +   ∩ −

closer a → 0 ↓ wider the parabola



ex   0.5   10

factorised form

$f(x) = a(x-p)(x-q)$ → sign of x

p, q → x-intersections

complete square

$f(x) = a(x-r)^2 + s$

vertex at (r, s)

↑ grades → ↑ wobbly

graph intersects x ≤ N° of grades

RATIONAL FUNCTIONS

$R(x) = \frac{p(x)}{q(x)}$ for polynomials $p(x)$, $q(x)$

If I want to express $R(x)$ as a function of $\frac{p(x)}{q(x)}$

then I need to have a common denominator

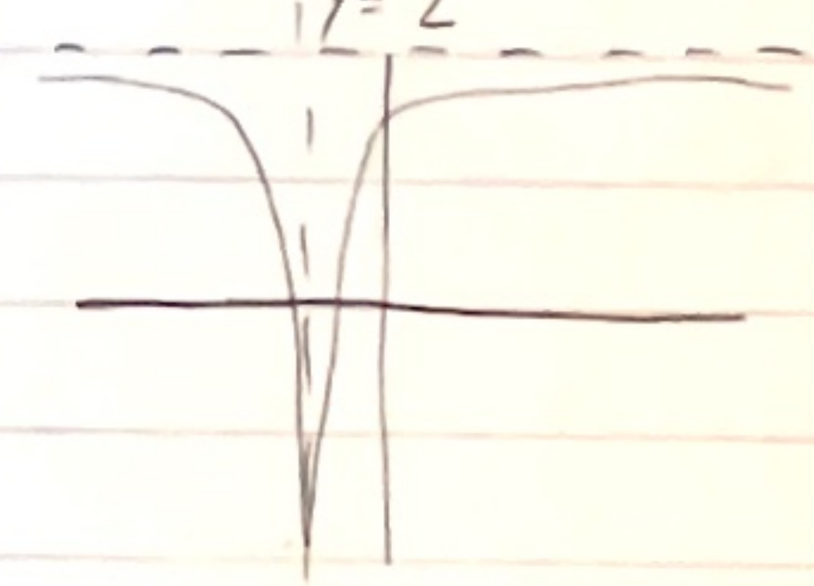
↓

Don't expand denominator unless really needed

$f(x) = \frac{2x^2 + 2x - 1}{x^2 + x + 1}$

Rational functions have asymptotes

horizontal asymptotes



tiene asíntota horizontal si el grado del numerador es al menos el grado del denominador

↑

⇒ has an horizontal asymptote if $\deg(p) \leq \deg(q)$

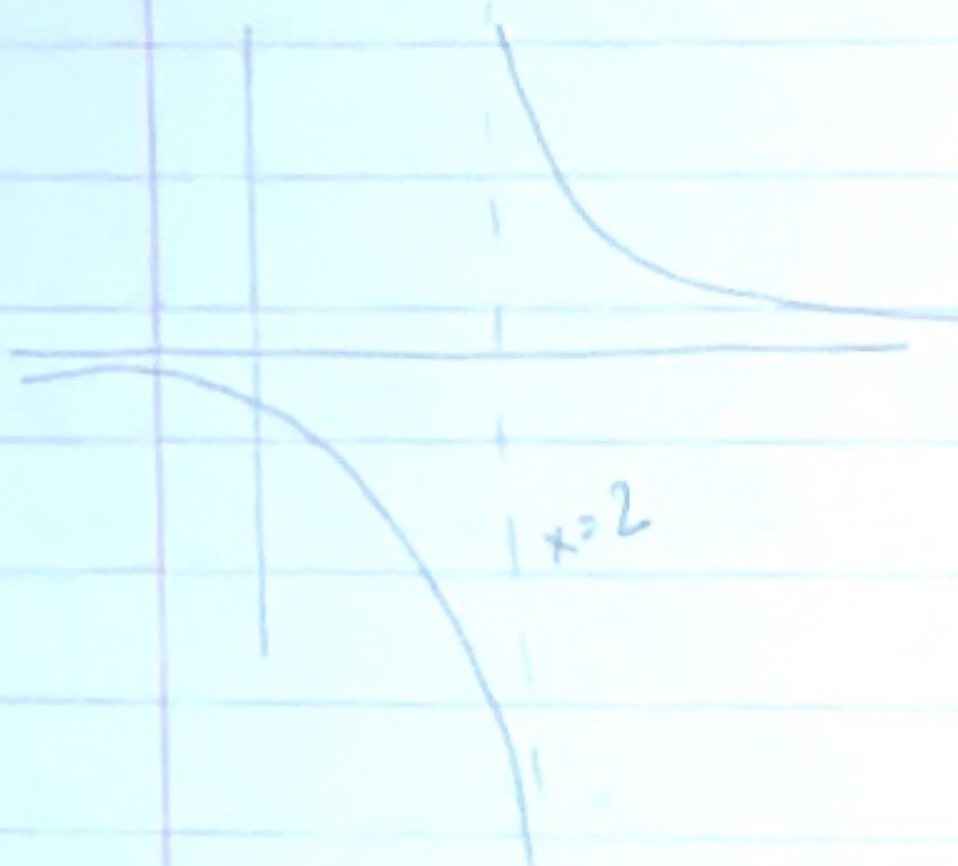
determine location by dividing by $x^{\deg(q)}$ → divide el numerador y denominador por x elevado al mayor grado de x en denominador

$$\frac{2x^2 + 2x - 1}{x^2 + x + 1} = \frac{2 + 2\frac{1}{x} - \frac{1}{x^2}}{1 + \frac{1}{x} + \frac{1}{x^2}} \approx \frac{2}{1} = 2$$

↑

for large x

Vertical asymptotes

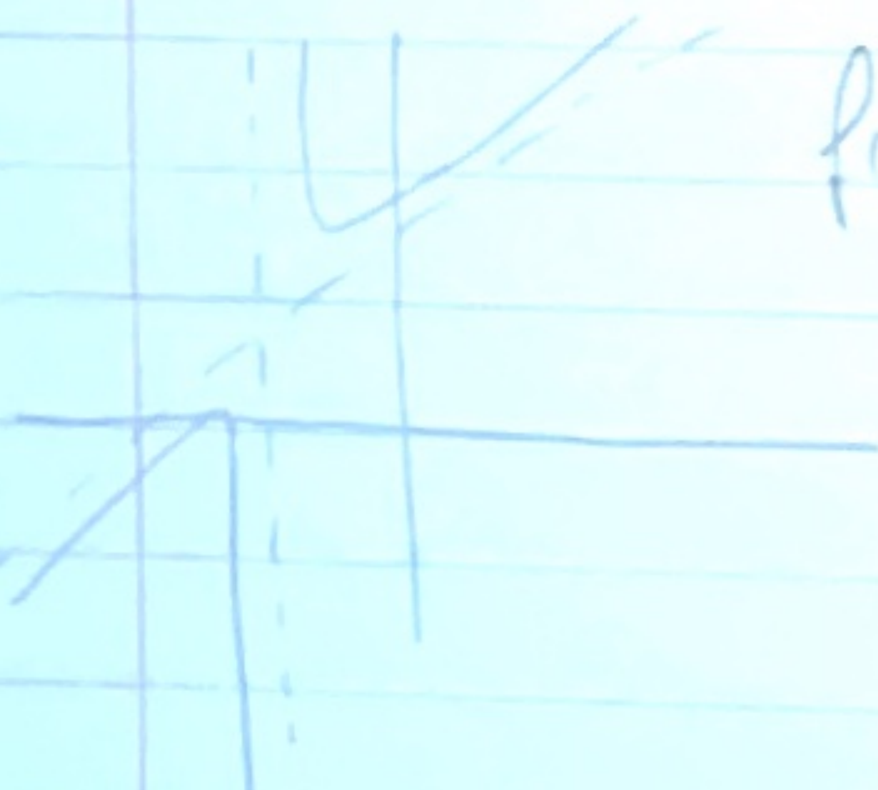


$f(x) = \frac{1}{x-2}$ — asymptote $x=2$

vertical asymptotes only if denominator $\to 0$ → faster

if numerator $= 0$ ⇒ then that number doesn't belong to domain

Oblique asymptotes



$f(x) = x + 2 + \frac{1}{x+1}$

$f(x) - (x+2) = \frac{1}{x+1}$

$x+2$ oblique asymptote

for large x ⇒ approximates to 0

Summary

$R(x) = \frac{P(x)}{Q(x)}$ for polynomials $P(x)$, $Q(x)$

Domain: excludes points with $Q(x) = 0$

Add: put over common denominator

horizontal asymptote: if $\deg(P) \leq \deg(Q)$ divide by $x^{\deg(Q)}$

vertical asymptote: only at $Q(x) = 0$

→ zeros wherever $P(x) = 0$ and $Q(x) \neq 0$

Power functions

tener cuidado con los dominios!

w/ integer exponents

$f(x) = x^a$ → where a is a constant

$f(x) = x^a$ for integer ≥ 0 → MONOMIAL

$x^a x^b = x^{a+b}$ $\qquad \dfrac{x^a}{b^a} = x^{a-b}$ $\qquad x, y > 0$

$(a^a)^b = x^{a \cdot b}$

$(xy)^a = x^a y^a$

x^2 parábola

x^3

↑ impares

x^4

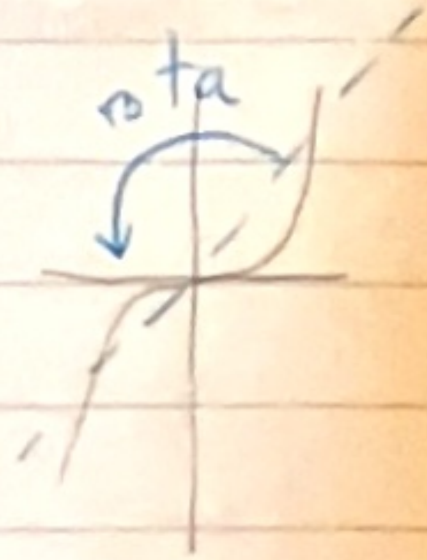
reflejo

↑ pares

even $f(-x) = f(x)$

x^5

rota



↑ odd

$f(-x) = -f(x)$

$f(x) = x^a \qquad a < 0$

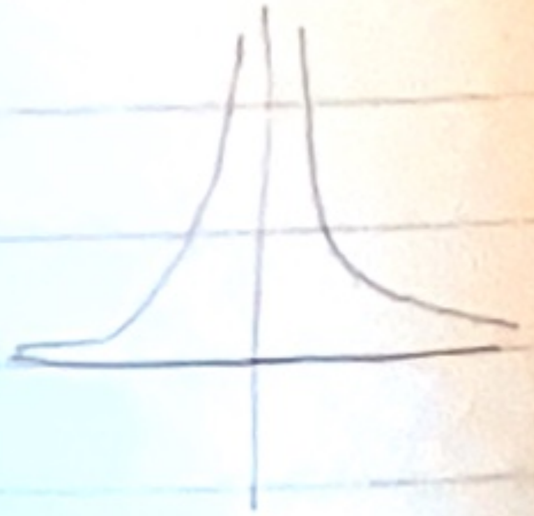
$x^{-1} = \dfrac{1}{x}$

hipérbola

impares

ej $x^{-3} = \dfrac{1}{x^3}$

$x^{-2} = \dfrac{1}{x^2}$



pares

graph has vertical asymptote at $x = 0$, $y = 0$

even → symmetrical in x axis

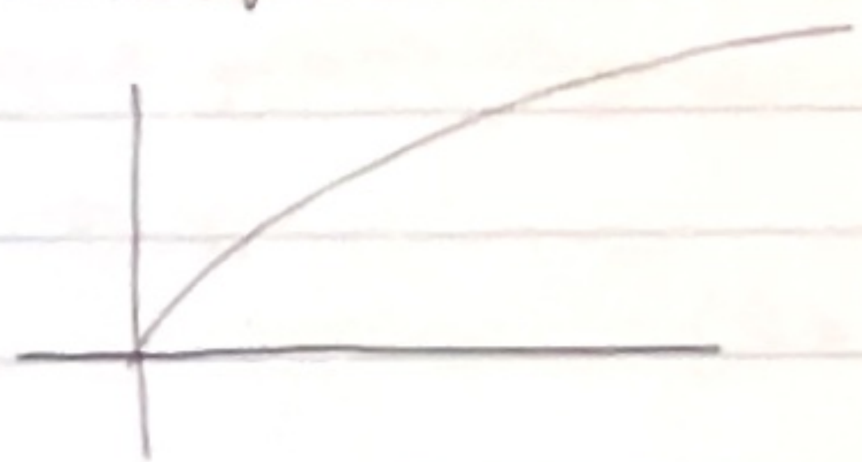
odd → rotan con eje en el origen

Non integer exponents

$f(x) = x^a \quad (x > 0)$

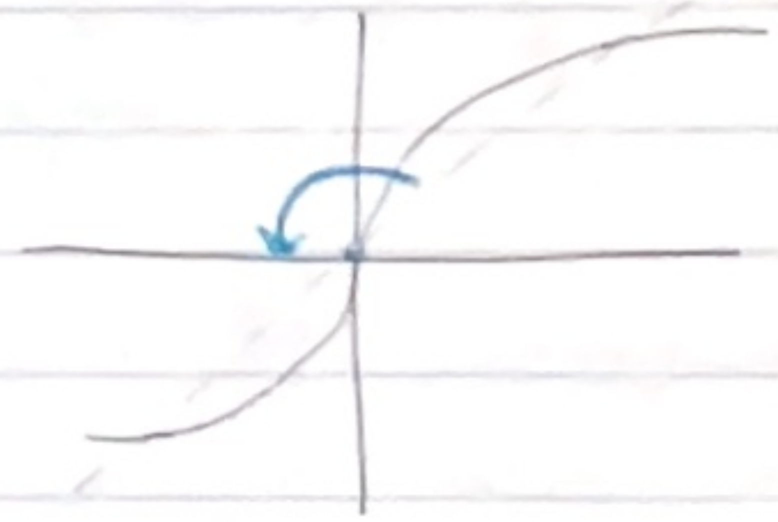
$a = \dfrac{1}{n}$ $\qquad \sqrt[n]{x}$

even values



domain $[0, \infty)$

odd



all $\mathbb{R}$

$x^{-5/2} = \left(\dfrac{1}{x^5}\right)^{1/2} = \dfrac{1}{\sqrt{x^5}} = \dfrac{1}{(\sqrt{x})^5}$

$x^{\frac{2}{3}} = (x^2)^{\frac{1}{3}} = \sqrt[3]{x^2} = (\sqrt[3]{x})^2$

$$x^{\frac{p}{q}} = \sqrt[q]{x^p} = \left(\sqrt[q]{x}\right)^p$$

$$x^{-\frac{p}{q}} = \frac{1}{\sqrt[q]{x^p}} = \frac{1}{\left(\sqrt[q]{x}\right)^p}$$

$f(x) = x^{\sqrt{2}}$ (?)

	$a \geq 0$	$a < 0$	
in general	$x \geq 0$	$x > 0$	domains
integer a or $\frac{p}{q}$ with q odd	all x	$x \neq 0$	

graphs $a > 0$ increasing for $x > 0$

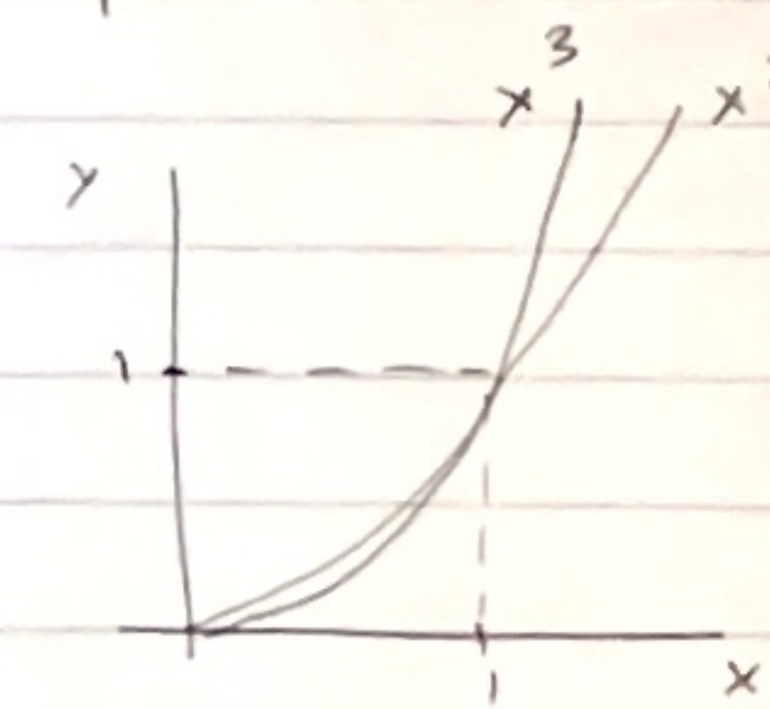
$a < 0$ decreasing for $x > 0$

asymptotes $x = 0$, $y = 0$

if defined for $x < 0$ either line symmetry in y axis or point symmetry in origin

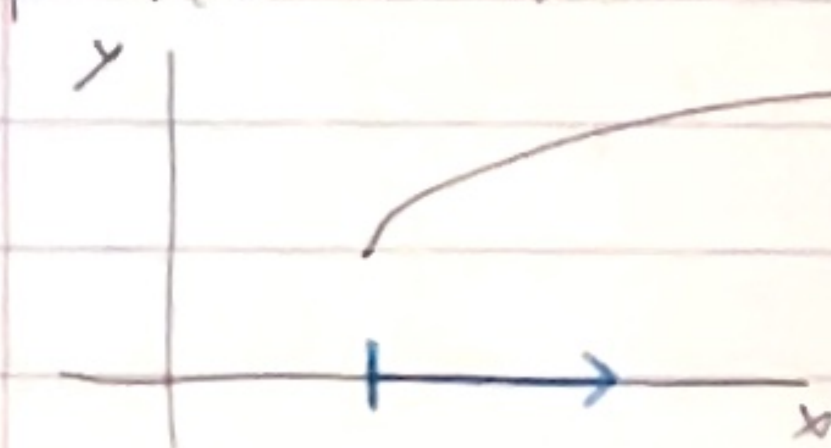
if $x > 1$ and $a < b \Rightarrow x^a < x^b$

if $0 < x < 1$ then $a < b$ then $x^a > x^b$



Domain, Codomain and Range

cf Maximal domain



$f(x) = 2 + \underbrace{\sqrt{x-3}}_{\geq 0}$

only condition $x \geq 3$

maximal domain : $[3, \infty)$

(we can choose)

$\mathbb{R}$ ← codomain = set containing all possible outputs

depends on domain ← range = set of outputs actually obtained

for which y does $f(x) = y$ have a solution

Domain → all inputs

Codomain → contain outputs

Range → attained outputs

Continuous functions

f is continuous at $x = a$

a small change on the x-axis gives a small change on the y-axis

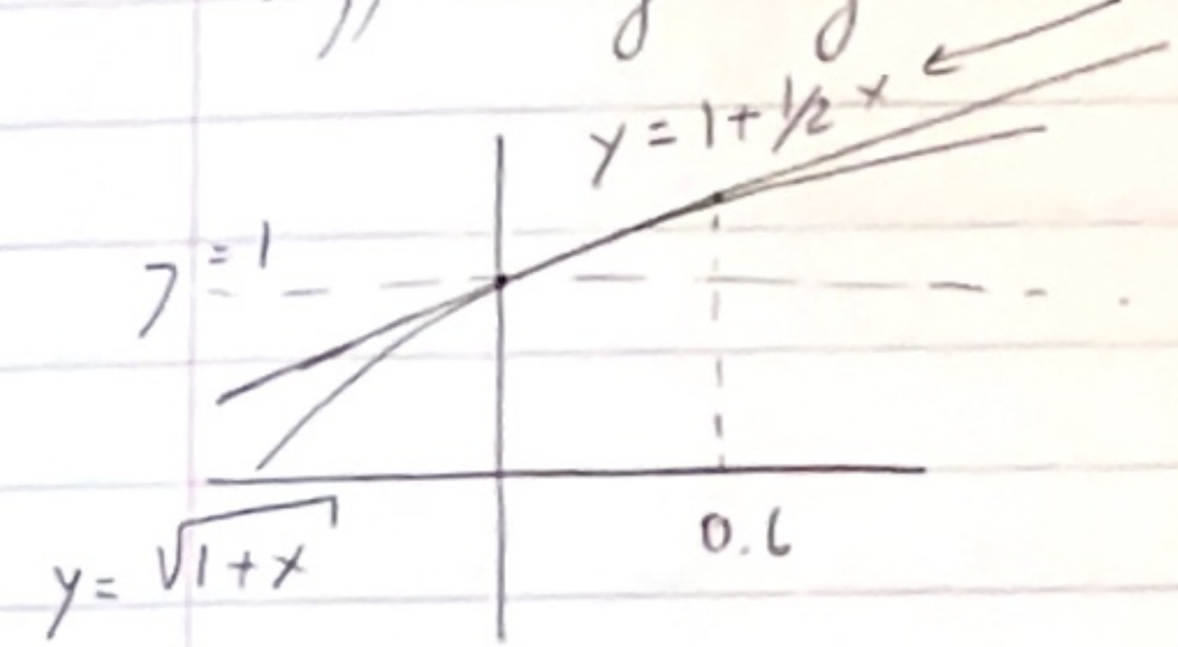
intermediate value property

if we combine 2 continuous functions then the product is continuous

! piecewise defined functions_

Taylor Polynomials

Approximating via a tangent



$y = 1 +$

there's 1 polynomial of degree 2 which best approximates $f(x) = \sqrt{1+x}$ near $x=0$

1) $1 + \frac{1}{2}x$ is a polynomial degree 1

2) we can approximate w/ a higher polynomial

2nd derivative of f = 2nd derivative of approximating polynomial

$$1 + \frac{x}{2} - \frac{x^2}{8}$$

↑ degree of polynomial ⇒ ↑ approximation